

Define Problem Statements

Field	Details
Team ID	SWTID-2026-7253
Team Size	5
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Project	OptiCrop: Smart Agricultural Production Optimization Engine

1. Problem Statement

Farmers often struggle to select the most suitable crop because soil nutrients and environmental conditions vary across regions. Traditional decisions rely on experience, leading to lower yields and inefficient resource utilization. An intelligent ML-based system is required to recommend suitable crops based on Nitrogen, Phosphorous, Potassium, temperature, humidity, pH and rainfall.

2. Existing System

Crop selection is mostly based on traditional knowledge, manual analysis and local experience. Decisions may not fully consider all environmental parameters.

3. Problems Identified in the Existing System

- Inefficient crop selection
- Low productivity
- Poor resource utilization
- No data-driven recommendations
- Difficult to analyze multiple factors
- Limited scalability

4. Root Cause Analysis

Root causes include lack of integrated soil analysis, dependence on manual decisions, insufficient environmental analysis and limited access to intelligent tools.

5. Proposed Solution

OptiCrop uses Machine Learning to analyze soil nutrients and climate parameters to recommend the most suitable crop and support sustainable farming.

6. Project Objectives

1. Recommend suitable crops.
2. Improve yield.
3. Optimize resource use.
4. Support sustainable agriculture.
5. Provide fast predictions.
6. Offer web-based access.

7. Project Scope

Includes soil data input, ML prediction, crop recommendation, environmental suitability assessment, reporting and analytics.

8. Expected Benefits

Higher productivity, improved decision-making, optimized fertilizer and water usage, sustainable farming, scalable web platform.

9. Conclusion

OptiCrop provides intelligent crop recommendations using AI and ML, helping farmers maximize productivity and resource efficiency while supporting sustainable agriculture.